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$$\vec{v}$$

$$(A - \lambda I) \vec{v}_1 = \vec{v}$$

$$\vdots$$

$$(A - \lambda I) \vec{v}_k = \vec{v}_{k-1}$$

Exercise:

1) Find all of the generalized e-vectors of the matrix

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 0 & 3 & -1 \\ 0 & 0 & 3 \end{bmatrix}$$

Since  $A$  is upper triangular,  $\lambda = 3$  ( $\lambda^3$ ).

Now as  $A - 3I = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$  is rank 2

we pick e-vector  $\vec{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ . Now

letting  $\vec{v}_1$  be such that  $\vec{v} = (A - 3I) \vec{v}_1$

we have that

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \vec{v}_1$$

and thus we pick  $\vec{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ . Finally

letting  $\vec{v}_2$  be such that  $\vec{v}_1 = (A - 3I)\vec{v}_2$

we have that

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \vec{v}_3$$

and thus  $(\vec{v}_3)_3 = -1$  and  $(\vec{v}_3)_2 = 2(\vec{v}_3)_3 = -2$

Now picking  $\vec{v}_3 = \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix}$  we have that

the set of generalized e-vectors is given by  $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix} \right\}$  which

are elements of  $\text{null}(A - 3I)$ ,  $\text{null}(A - 3I)^2$  and  $\text{null}(A - 3I)^3$ . Hence  $\mu_A = (x - 3)^3$ .

Theorem 18.1  $A \in \mathbb{C}^{n \times n}$  is diagonalizable, i.e.

$A$  has  $n$  linearly independent e-vectors,  
iff its minimal polynomial has only simple  
(order 1) zeros.

Proof: let  $\mu = \prod_{i=1}^k (x - \lambda_i)$  such that  $\lambda_i \neq \lambda_j$

and set  $\frac{1}{\mu} = \sum_{i=1}^k \frac{\alpha_i}{x - \lambda_i}$ . Then  $1 = \sum_{i=1}^k \frac{\alpha_i \mu(x)}{x - \lambda_i}$

and thus

$$\begin{aligned} x - \lambda_j &= (x - \lambda_j) \sum_i \frac{\alpha_i \mu(x)}{x - \lambda_i} \\ &= \alpha_j \mu(x) + \sum_{i \neq j} \frac{\alpha_i \mu(x) (x - \lambda_j)}{x - \lambda_i} \\ &= \alpha_j \mu(x) + (x - \lambda_j)^2 p(x) \end{aligned}$$

Now differentiating both sides we get

$$1 = \alpha_j \mu'(x) + (x - \lambda_j) q(x)$$

and thus setting  $x = \lambda_j$  we have that

$$\alpha_j = \frac{1}{\mu'(\lambda_j)} \quad \text{and thus} \quad \frac{1}{\mu} = \sum_i \frac{1}{\mu'(\lambda_i)} \frac{1}{x - \lambda_i}.$$

Now setting  $g_i(x) = \frac{\mu(x)}{\mu'(\lambda_i)(x - \lambda_i)}$  so that  $\sum_i g_i(x) = 1$

and thus  $\sum g_i(A) = I$ . Since  $(x - \lambda_i) g_i = \frac{\mu}{\mu'(\lambda_i)}$

we have that  $(A - \lambda_i I) g_i(A) \cdot \vec{x} = \frac{\mu(A)}{\mu'(\lambda_i)} \cdot \vec{x}$

$$= \vec{0}$$

and thus  $q_i(A) \cdot \vec{x} \in \text{null}(A - \lambda_i I)$  for  $i = 1, \dots, k$ . Now as  $\vec{x} = \sum_i q_i(A) \cdot \vec{x}$  for

any  $\vec{x} \in \mathbb{C}^n$  and  $\text{null}(A - \lambda_i I) \cap \text{null}(A - \lambda_j I) = \emptyset$  we have that  $\mathbb{C}^n = \bigoplus_{i=1}^k \text{null}(A - \lambda_i I)$ .

Now we turn our attention to matrices that are not diagonalizable.

## Jordan Normal Form

Def 18.2 let  $T \in \mathcal{L}(V, V)$  and let  $W$  be a subspace of  $V$ . Then  $T|_W$  is the restriction of  $T$  to  $W$ . I.e. for each  $\vec{x} \in W$   $T|_W(\vec{x}) = T(\vec{x})$ .

Theorem 18.3 let  $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$  and  $A = [T]_{\{\vec{e}_i\}}$  be such that for non-zero vectors  $\vec{v}_i, \vec{v}_{i_1}, \dots, \vec{v}_{i_m}$

Jordan chain for  $\vec{v}_i$  and has length  $m+1$

$$\begin{aligned} A \vec{v}_i &= \lambda_i \vec{v}_i \quad \text{or} \quad (A - \lambda_i I) \vec{v}_i = \vec{0} \\ (A - \lambda_i I) \vec{v}_{i_1} &= \vec{v}_i \\ (A - \lambda_i I) \vec{v}_{i_2} &= \vec{v}_{i_1} \\ &\vdots \\ (A - \lambda_i I) \vec{v}_{i_m} &= \vec{v}_{i_{m-1}} \end{aligned}$$

Then for  $W = \text{span}(\vec{v}_i, \vec{v}_{i_1}, \dots, \vec{v}_{i_m})$

we have that

$$[T|_W]_{\{\vec{v}_{i_k}\}} = \begin{bmatrix} \lambda_i & \color{red}{1} & 0 & \dots & 0 \\ 0 & \lambda_i & \color{red}{1} & \dots & 0 \\ 0 & 0 & \lambda_i & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_i \end{bmatrix}_{(m+1) \times (m+1)}$$

$T\vec{v}_i = \lambda_i \vec{v}_i$   
 $T\vec{v}_{i_1} = \vec{v}_i + \lambda_i \vec{v}_{i_1}$   
 $\vdots$   
 $T\vec{v}_{i_k} = \vec{v}_{i_{k-1}} + \lambda_i \vec{v}_{i_k}$

which is called a Jordan block matrix.

Proof: As  $(A - \lambda_i I) \vec{v}_{i_k} = \vec{v}_{i_{k-1}}$  for  $2 \leq k \leq m$

we have that  $A \vec{v}_{i_k} - \lambda_i \vec{v}_{i_k} = \vec{v}_{i_{k-1}}$

and thus  $A \vec{v}_{i_k} = \vec{v}_{i_{k-1}} + \lambda_i \vec{v}_{i_k}$ . Now

the  $k^{\text{th}}$  column of  $[T|_W]_{\{\vec{v}_{i_k}\}}$  is

$$\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \\ \lambda_i \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} k-1 \\ k \end{matrix}$$

Def 18.4 Let  $\vec{v}_i$  be an e-vector of  $A$  with corresponding e-value of  $\lambda_i$  and let  $\{\vec{v}_i, \vec{v}_i, \dots, \vec{v}_i, \dots, \vec{v}_i\}$  be the associated Jordan chain of length  $n_i$ . Then the Jordan block matrix associated to  $\vec{v}_i$  is the  $n_i \times n_i$  matrix

$$J_i = J_{\lambda_i, n_i} = \begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}_{n_i \times n_i}$$

Theorem 18.5 Let  $A \in \mathbb{C}^{n \times n}$  have e-vectors of  $\vec{v}_1, \dots, \vec{v}_l$  ( $l \leq n$ ) with corresponding e-values of  $\lambda_1, \dots, \lambda_l$ . Then  $A = M J M^{-1}$ , i.e.  $A \sim J$ , where  $J$  is the block diagonal matrix

$$J = J_1 \oplus \dots \oplus J_l$$

$$= J_{\lambda_1, n_1} \oplus \dots \oplus J_{\lambda_l, n_l}$$

and  $M$  consists of all generalized e-vectors

(i.e. all Jordan chains). Here we say that  $J^*$  is the Jordan normal form of

\* Some use  $A = MJM^{-1}$  as the Jordan form

Example: The Jordan normal form of

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 0 & 3 & -1 \\ 0 & 0 & 3 \end{bmatrix} \text{ is } J = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix} \text{ or}$$

$$A = M \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix} M^{-1} \text{ where } M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & -1 \end{bmatrix}$$

Exercise:

2) Find the Jordan normal form of the matrix

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 0 & 4 & 0 \\ 1 & -2 & 5 \end{bmatrix}$$

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 3-\lambda & 2 & -1 \\ 0 & 4-\lambda & 0 \\ 1 & -2 & 5-\lambda \end{vmatrix} = (4-\lambda) \begin{vmatrix} 3-\lambda & -1 \\ 1 & 5-\lambda \end{vmatrix} \\ &= (4-\lambda)(\lambda^2 - 8\lambda + 16) \\ &= (4-\lambda)^3 \\ \lambda &= 4 \text{ (x3)} \end{aligned}$$

$$\text{Set } (A - 4I) \vec{x} = \vec{0} :$$

$$\left[ \begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 \end{array} \right]$$

$$x_1 = 2x_2 - x_3$$

$$x_2 = x_2$$

$$x_3 = x_3$$

$$\text{we pick } \vec{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \text{ and } \vec{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Now looking at the Jordan chains

$$\vec{v}_1 : \text{ setting } (A - 4I) \vec{v}_{1,1} = \vec{v}_1$$

$$\begin{bmatrix} -1 & 2 & -1 \\ 0 & 0 & 0 \\ 1 & -2 & 1 \end{bmatrix} \vec{v}_{1,1} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$\vec{v}_1$  length 1

$\vec{v}_2$  length 2

we obtain no solutions

$$\vec{v}_2 : \text{ setting } (A - 4I) \vec{v}_{2,1} = \vec{v}_2$$

$$\begin{bmatrix} -1 & 2 & -1 \\ 0 & 0 & 0 \\ 1 & -2 & 1 \end{bmatrix} \vec{v}_{2,1} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

pick  $\vec{v}_{2,1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Now  $A = M J M^{-1}$  where  $J = \begin{bmatrix} [4] & & \\ & \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix} & \\ & & \end{bmatrix}$

$$= \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{bmatrix}$$

and  $M = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

I would say the Jordan normal form  
of  $A$  is  $J = \begin{bmatrix} 4 & & \\ & 4 & 1 \\ & & 4 \end{bmatrix}$